OKBA BOUZIANE

Full-Stack & AI Engineer

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TECHNICAL SKILLS

AI & LLM: LangGraph, LangChain, LlamaIndex, AI Agents, RAG, Graph RAG, MCP, PGVector, Prompt Engineering, LLM APIs
Frontend: Angular, React, TypeScript, RxJS, React Native
Backend: Node.js, NestJS, Express.js, Python, FastAPI, REST, GraphQL, gRPC
Databases: PostgreSQL, MongoDB, Neo4j, Redis, MinIO (S3)
Security & DevOps: Keycloak, OAuth2 / OpenID Connect, RBAC, Docker, CI/CD, Git
AI Tools: Claude Code, Cursor, Windsurf, Kiro

PERSONAL PROJECTS

Secure Agent Workspace | React, NestJS, LangGraph, Keycloak, PostgreSQL
An AI assistant that works on company data with the permissions of the user who is logged in.

- The agent can only do what the user is allowed to do, and every change waits for a human to approve it.
- Answers questions from company documents with sources, blocks prompt injection, and has 82 automated tests.

VoiceDesk | Python, FastAPI, LiveKit, LangGraph, gRPC, PostgreSQL
A voice receptionist you talk to in the browser, live online.

- Books, lists and cancels appointments by voice, and answers questions from a knowledge base.
- Hands the call to a human when needed; bookings run in a separate gRPC microservice.

LexGraph | Python, FastAPI, LlamaIndex, Neo4j, LangGraph
A contract search assistant that follows the links between documents.

- Stores contracts, amendments and guarantees as a Neo4j knowledge graph to answer multi-document questions.
- On those questions it finds 92% of the needed documents versus 74% for plain vector search; covered by 41 automated tests.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Smartest Algeria, Algiers — Nov 2022 – Oct 2026

- Operations teams were reviewing documents and filling forms by hand inside the business applications; integrated LLM APIs for document summarization, smart search and form auto-fill, built end to end with Angular, React and NestJS.
- Each product had its own login and duplicated access rules; integrated Keycloak (OAuth2/OpenID Connect) with JWT-secured APIs and Role-Based Access Control for single sign-on across the applications, and built a custom admin console on the Keycloak Admin REST API for users, roles and clients.
- Licensing was handled separately in each application; designed and built a SaaS-style licensing platform that every Smartest application integrates with, centralizing license issuing and validation in one service.
- Business users depended on developers for every new report; built a Word-like report designer in which users bind data sources and datasets to tables and charts on a document canvas and generate reports themselves.
- Several applications needed visual flow editing, and each would otherwise have rebuilt graph rendering from scratch; designed flow-dc, a reusable Angular flow-graph editor library integrated across the apps, with an extension API that lets developers register custom node and edge types without modifying the core.
- Legacy modules were tightly coupled and risky to change; refactored them toward Clean Architecture with unit tests and a shared Angular component library, optimized PostgreSQL and MongoDB queries behind slow reporting screens, integrated MinIO (S3-compatible) for file storage, and shipped services with Docker through CI/CD.

Mobile & Web Developer | Tebessa Innoest, Tebessa — Mar 2020 – Jan 2022

- Incubated founders had ideas but no product to pitch; turned them into mobile and web MVPs in short delivery cycles, advising on technical scope and feature priority so each team could demo a working first version.
- Start-ups needed apps on both platforms on a small budget; built cross-platform React Native apps (iOS and Android) with authentication, REST APIs, push notifications and offline-friendly screens, plus React websites for demo days.

EDUCATION

MSc, Artificial Intelligence | Badji Mokhtar University, Annaba — 2020 – 2022
BSc, Computer Science | Badji Mokhtar University, Annaba — 2017 – 2020

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Full Professional), French (Limited Working)
Military Service: Exempt